

Bias Analysis by Gender

Gender: Female
Accuracy: 0.85

Gender: Male
Accuracy: 0.83

Bias Analysis by Senior Citizen Status

Senior Citizen: 0
Accuracy: 0.85

Senior Citizen: 1
Accuracy: 0.79

Observation:

- Slightly lower accuracy observed for senior citizens.
- No major bias detected between male and female groups.

Confusion Matrix

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[[952 83]
 [145 229]]
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Top Feature Importances

	Feature	Importance
3	tenure	0.243
5	MonthlyCharges	0.182
8	Contract	0.145
10	InternetService	0.121
15	PaymentMethod	0.098
4	TotalCharges	0.076
6	OnlineSecurity	0.041
7	TechSupport	0.031
2	Partner	0.028
1	gender	0.020

Generating SHAP Summary Plot...

[SHAP Summary Plot Displayed]

- Features such as tenure and monthly charges showed the highest impact on churn prediction.
- High monthly charges increased churn probability.
- Long tenure reduced churn probability.

Generating SHAP Force Plot...

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